

Elementary Matrices: Construction and Application

High-Level Overview

In the study of linear algebra and computational mathematics, we often need to simplify matrices to solve systems of linear equations or compute inverses. This simplification is achieved through **elementary operations**.

An **elementary matrix** is a special matrix created by performing a single elementary operation on an identity matrix (I). They act as mathematical "switches" or "transformations":

- **Pre-multiplying** (multiplying on the left, EA) performs **row operations** on matrix A .
 - **Post-multiplying** (multiplying on the right, AE) performs **column operations** on matrix A .
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1. What is an Elementary Matrix?

An **elementary matrix** is obtained from a unit/identity matrix by subjecting it to any one of the elementary transformations.

Examples of Elementary Matrices:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Fundamental Theorem of Elementary Transformations

Elementary row (or column) transformations of a matrix A are obtained by pre-multiplying (or post-multiplying) A by the corresponding elementary matrices.

Demonstration: Left vs. Right Multiplication

$$\text{Let } E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

- **Left Multiplication (EA - Row Operation):**

$$EA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$$

◦ *Effect:* Left multiplication by E interchanges the first and second rows of A .

- **Right Multiplication (AE - Column Operation):**

$$AE = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 5 & 4 & 6 \\ 8 & 7 & 9 \end{bmatrix}$$

◦ *Effect:* Right multiplication by E interchanges the first and second columns of A .

3. Construction of Elementary Matrices

An elementary matrix E of order $n \times n$ that performs a specified elementary row operation is obtained by performing that exact same operation on an identity matrix of equal order.

Worked Examples with a 3×4 Matrix

Let us consider the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$$

Case 1: Interchanging the first and third rows of A

1. Start with a 3×3 identity matrix I :

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Apply the operation (interchanging row 1 and row 3) to I to get the elementary matrix E :

$$E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

3. Multiply E by A :

$$EA = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 9 & 10 & 11 & 12 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

Case 2: Subtracting 5 times the first row from the second row of A ($R_2 \rightarrow R_2 - 5R_1$)

1. Start with the 3×3 identity matrix I and apply the row operation:

$$R_2 \rightarrow R_2 - 5R_1 \text{ on } I \implies E = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Multiply E by A :

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -4 & -8 & -12 \\ 9 & 10 & 11 & 12 \end{bmatrix}$$

Case 3: Interchanging the first and third columns of A

*Note: Since we need to perform a **column operation** on a 3×4 matrix, we must right-multiply by a 4×4 elementary matrix.*

1. Start with a 4×4 identity matrix I :

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2. Apply the column operation ($C_1 \leftrightarrow C_3$) to I to construct E :

$$E = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3. Post-multiply A by E :

$$AE = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 & 4 \\ 7 & 6 & 5 & 8 \\ 11 & 10 & 9 & 12 \end{bmatrix}$$

Quick Exam Revision Summary

- **Definition:** An elementary matrix is created by applying a single row or column operation to an identity matrix.
- **Row Operations (Pre-multiplication):** Multiplying EA transforms the **rows** of A . Construct E by applying the row operation to an identity matrix matching the rows of A .
- **Column Operations (Post-multiplication):** Multiplying AE transforms the **columns** of A . Construct E by applying the column operation to an identity matrix matching the columns of A .